



CETM46

Data Science Product Development

**Building a Telecom Churn Dashboard: A Data Science Approach
to Customer Retention**

Assignment 2

Submitted By:

Student Name: Riza Khati

Student ID: 239419172

Submitted To:

Ming Jiang

Module Leader

Table of Contents

1. Introduction	3
2. Product Design Section.....	3
2.1. Application Domain/ End User's Requirements Analysis.....	3
2.2. Data Source and Theme Selection and Specification	3
2.3. Product Functional and Non-Functional Requirement Specifications.....	3
2.3.1. Functional Requirements	3
2.3.2. Non-functional Requirements	4
2.4. Product Software Architecture Design	4
2.5. Product Use Case Specifications	5
3. Product Development Section	7
3.1. Selection of Appropriate Software Tools/Platforms and Hardware Methodologies	7
3.2. Product Development Software Engineering Methodology	7
3.3. System Testing Method	8
3.4. User Evaluation Plan and Methods.....	8
4. Project Management Section	9
4.1. Time Management with Gantt Chart.....	9
4.2. Risk Assessment on Personal Information Protection and Data Security/Government.....	9
4.3. Quality Control on Software Development.....	9
4.4. Basic Customer/User Relationship management	9
4.5. Basic Product Marketing Strategy	10
5. Conclusion.....	11
6. References	12
7. Appendix	13

1. Introduction

For telecom enterprises, customer attrition is a serious issue leading to diminished profitability (Pejić Bach, et al., 2021). The global telecommunications industry is highly competitive, making it increasingly difficult to retain customers in the face of market saturation. Even with large marketing budgets spent on attracting new clients, maintaining an existing clientele is usually more cost-effective. Research indicates that a company's profits can rise by 25% to 85% when its turnover rate is lowered by 5% (Yang, 2023). As a result, reducing customer attrition has become a top priority for telecom companies (Zhang, et al., 2022).

The conceptualization of the Telecom Churn Analysis Dashboard emerged from recognizing significant challenges within the telecommunications sector. This report critically evaluates the development process of a scientific product tailored to address these challenges, encompassing project management, software development, and user interface design. Python is the primary tool for data processing, modeling, and analysis. The objective is to create an interactive dashboard, "Telecom Churn Dashboard," to facilitate informed decision-making through visualizing contractual impacts, enhancing customer profiling, predictive modeling, and demystifying churn patterns. The dataset sourced from Kaggle, the project entails comprehensive stages including data cleaning, exploratory data analysis (EDA), feature analysis, and identifying important features for modeling during the experimental phase.

2. Product Design Section

2.1.Application Domain/ End User's Requirements Analysis

As information technology advances, the volume of data has surged, enabling organizations to store and process large amounts of information, fostering the automatic creation of knowledge. Data mining and machine learning are increasingly utilized to analyze this data, particularly in customer relationship management (CRM) for subscription-based businesses. Understanding customer behavior through data analysis is crucial for predicting and preventing customer churn, which is essential for maintaining stable revenue streams and minimizing acquisition costs (Zimal, et al., 2023).

Developing a customer risk analysis dashboard involves several stages: data collection, preprocessing, model training, validation, and evaluation. The effective use of training and testing data streamlines the process, ensuring the accuracy of the model.

2.2.Data Source and Theme Selection and Specification

The data used in this report is sourced from the IBM Telcom dataset available on the Kaggle [URL](#), focusing on customer data and program retention. It includes details such as customer personal information, program-related services, churn actions, and demographic data. The dataset consists of 21 features and covers a total of 7034 customers. Within the IBM Telcom dataset, approximately 26.54% of the samples represent churn cases, and 73.4% of customers do not churn.

2.3.Product Functional and Non-Functional Requirement Specifications

2.3.1. Functional Requirements

- Pre-process and clean data to analyze the churn.
- Identify the important features influencing customer churn.
- Develop an effective prediction model.

- Analyse customer data should be reliable and accurate.
- Create an interactive dashboard to visualize churn patterns and predictive outcomes.

2.3.2. Non-functional Requirements

- The interface effortlessly clarifies customer churn patterns via interactive visualizations.
- Users can get reports and visualizations of the predicted churn customer outcomes.
- The product can adapt to business trends.
- Use filters to evaluate how certain campaigns, customer segments, and contract terms influence churn behaviors.
- Save time and reduce operating costs by utilizing available data instead of relying on outdated manual methods.
- User data must be protected with data regulation.

2.4. Product Software Architecture Design

The system is designed to distinguish between customers who churned and those who stayed and to determine factors that contribute to churn. First, data cleaning, data transformation, and data normalization are carried out in the pre-processing process. The product was developed using Python Libraries such as Pandas, and Numpy for analyzing and calculating data and is essential for implementing the array data type and exploratory data analysis (EDA). It is mainly used for matrix calculations. This is followed by the EDA step its purpose is to make sense of the data and help us better understand each feature before feeding them to the machine learning models, thereby, making the modeling more efficient.

Secondly, machine learning classifiers are used to predict customer churn Logistic regression model. Besides, the oversampling method SMOTE is applied to the training set to improve the classifier's performance. Finally, based on the results of churn prediction and customer segmentation, customer behaviour analytics are conducted. The uncertainty of churn is visualized by Power BI. Different machine learning classifiers are compared. The characteristics of each churn segment are summarised, which can be useful for marketers or strategists to come up with different retention measures for different segments (Saha, et al., 2024).

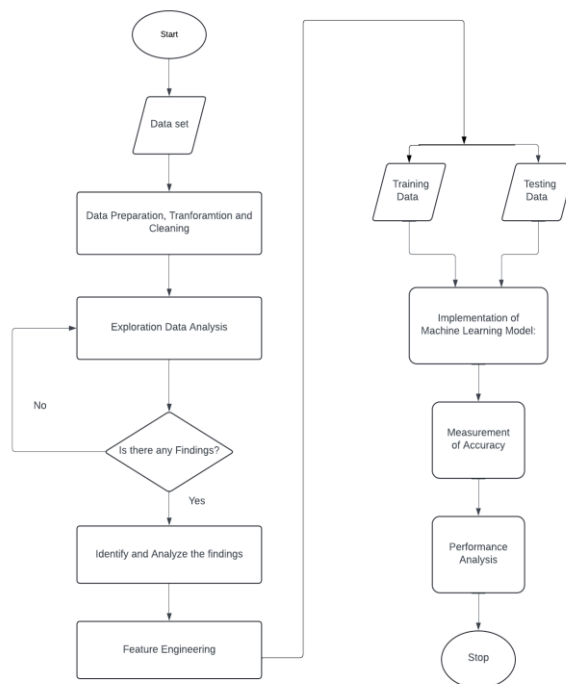


Figure 1: Purposed Software Architecture

2.5.Product Use Case Specifications

This approach establishes a robust and precise risk analysis system tailored for stakeholders.

Use Case: Customer Risk Analysis

User: CRM manager/ Customer retention analyst

Scenario: A customer retention analyst needs to predict which customers are at risk of leaving the telecom service.

Action: The user should select the group they want to focus on by choosing parameters such as gender, contract type, and subscription type. The user then accesses the churn risk analysis dashboard to view metrics, including usage patterns, with visualizations. The dashboard also provides churn rates and churn probabilities for customers based on predictive analytics models.

Outcome: The analyst identifies high-risk customers and develops targeted retention strategies, such as personalized offers or proactive outreach, to reduce churn rates.

Use Case: Telecom Churn Dashboard

User: CRM manager/ Customer retention analyst

Scenario: A customer retention analyst at a telecom company needs to identify and mitigate the risk of customer churn by analyzing different segments of the customer base. The analyst aims to use the churn dashboard to understand the churn risk across various demographics, account details, and subscription types.

Action: The dashboard presents various visualizations, including Charts and graphs showing the distribution of customers by gender, age group, and region. Visualizations such as bar charts and pie charts displaying the proportions of customers with different contract types and account tenures. Percentage showing the distribution of customers across different subscription plans.

Outcomes: The customer retention analyst effectively identifies high-risk customer segments through demographic, account, and subscription data visualizations.

Use Case: Revenue

User: CRM manager/ Customer retention analyst

Scenario: The analyst uses the churn dashboard to analyze customer charges and identify high-risk customers, aiming to take proactive measures to retain these customers and minimize revenue impact.

Action: The dashboard presents various visualizations, including:

Monthly charge insight: The visualizations indicate that customers with higher monthly charges (e.g., subscribers) are at greater risk of churn.

Total Charges Insight: Customers who have accumulated significant total charges over time but exhibit high churn probabilities represent a critical revenue risk.

Outcome: The revenue analyst effectively identifies high-risk customer segments contributing significantly to revenue through monthly and total charge visualizations.

3. Product Development Section

3.1. Selection of Appropriate Software Tools/Platforms and Hardware Methodologies

This section shows and describes the choice of the different programming languages, libraries, and frameworks used in building the application. To design the data dashboard for customer churn analysis. While there exist several tools capable of fulfilling this purpose, such as Elasticsearch and IBM Cognos, after evaluating both functional and non-functional requirements, the most cost-effective and straightforward solution appears to be Power BI. Power BI offers a user-friendly interface and facilitates rapid development of intuitive data dashboards. Moreover, it provides extensive customization options and is built to meet all project requirements. By leveraging Power BI, the development process becomes more accessible and economical (Cybiant, 2020).

Additionally, complementing Power BI with Python ensures enhanced flexibility and analytical capabilities. Critically evaluate the suitability of the chosen technologies for implementing key functionalities of the product. Its straightforward syntax and extensive library support, particularly in data analytics and machine learning, made it the ideal choice for this project (Kolomiets, et al., 2021).

Data analytics and machine learning libraries: Matplotlib, Sklearn, NumPy, Pandas, and Seaborn are key Python libraries used in our project. Matplotlib creates plots and graphs, while Sklearn offers machine-learning algorithms and preprocessing tools. NumPy supports multidimensional arrays, Pandas aids data wrangling, and Seaborn provides statistical visualization.

3.2. Product Development Software Engineering Methodology

The project adopted a waterfall methodology to develop the product user requirements for planning, documentation, and sequential progress through predefined phases, with limited flexibility for changes once the project is underway. Each phase must be finished before progressing to the next one. These phases encompass gathering requirements, designing, developing, testing, deploying, and maintaining the project.



Figure 2: Waterfall Model

3.3. System Testing Method

I initiated the testing section by evaluating each model's performance to ensure it meets the end user's needs. This assessment took place within Jupyter Notebook, our experimental development environment. Utilizing accuracy and classification metrics like confusion matrices, I compared the performance of logistic regression, random forest classifiers, Decision trees, naive Bayes, and KNN. The results, showcased in the figure, reveal that while all models achieved similar accuracy rates, logistic regression displayed slightly better performance in precision, recall, F1-score, and ROC curve. Therefore, logistic regression appears to be the preferable model for this scenario based on these metrics.

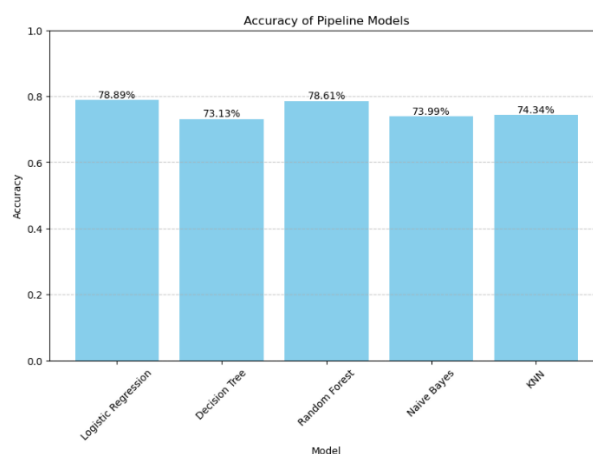


Figure 3: Models Accuracy Rates

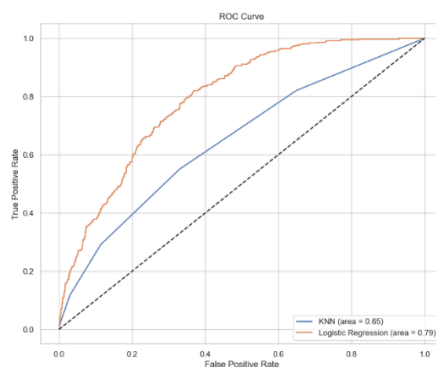


Figure 4: ROC Curve for Logistic and Random Forest

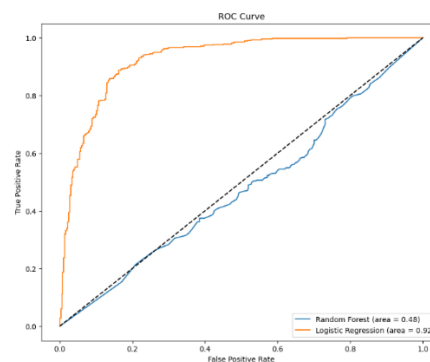


Figure 5: ROC curve for KNN and Logistic Model

3.4. User Evaluation Plan and Methods

At the end of development, analyze the product's alignment with functional and non-functional requirements. A thorough examination determines whether the product has adequately fulfilled the stipulated criteria, thereby indicating its completion to the requisite standard. Additionally, adherence to a set of quality assurance measures will be ensured to guarantee the absence of bugs, crashes, or any performance issues that might impede user experience.

4. Project Management Section

4.1.Time Management with Gantt Chart

During a two-month project, an Excel-generated Gantt chart was utilized to track the product's development, assigning specific dates to each task to ensure successful outcomes. Efficient time management is pivotal in achieving success in customer churn prediction development. This structured timeline facilitated adherence to the project's goals and enabled the effective implementation of the chosen methodology.

4.2.Risk Assessment on Personal Information Protection and Data Security/Government

An essential aspect of managing telecom customer churn data involves addressing the potential breach of confidentiality. Unauthorized access to such data is unfair as it confers advantages to individuals with privileged information. Nevertheless, this concern is mitigated on platforms like Kaggle, where telecom data is limited to publicly available information.

This abundance of data doesn't always translate to meaningful information. There's a risk of misuse of data mining techniques leading to the unauthorized exposure of sensitive user information. This has caused growing concern among people, with many hesitant to share their real data. Therefore, it's crucial to explore how data mining can be used to protect personal privacy and analyze data effectively. According to the GDPR Data Protection Act, individuals have the right to access, correct, delete, or limit their data processing. However, data must be utilized fairly, legally, and transparently, serving specific and explicit purposes. It should be relevant, limited to necessary usage, accurate, and updated when required. Additionally, data should only be retained for as long as necessary and handled securely to prevent unauthorized access, loss, or damage (Anon., n.d.).

4.3.Quality Control on Software Development

In the quality control process for developing a customer churn prediction dashboard, we leverage Python within a Jupyter Notebook environment for Exploratory Data Analysis (EDA) using Pandas and NumPy. Pandas facilitate data manipulation and preprocessing tasks, ensuring data cleanliness and consistency, while NumPy supports numerical operations for data computations. Matplotlib is utilized to generate comprehensive visualizations, aiding in identifying patterns and trends in customer data relevant to churn prediction. These insights guide the development and validation of machine learning models. Upon preprocessing and EDA, the processed data is seamlessly exported to Power BI for advanced visualization and interactive dashboard creation, enabling stakeholders to conduct in-depth customer risk analysis. By employing this integrated approach, we ensure the churn prediction dashboard's accuracy, reliability, and usability, thus empowering businesses to make proactive decisions to mitigate customer churn and enhance retention strategies.

4.4.Basic Customer/User Relationship management

The primary challenge lies in how telecommunication service providers can transition from conventional sales-focused customer relationship management (CRM) systems to advanced analytics CRM platforms, aimed at integrating insights from customer interactions and transaction processing to mitigate customer churn. CRM entails marketing and customer service strategies geared towards fostering enduring relationships with customers. By identifying repeat customers and analyzing their behavior, organizations glean valuable insights to enhance conversion rates. Hence, CRM serves as a pivotal tool for cultivating improved customer relationships within an organization (Ledro, et al., 2023).

4.5.Basic Product Marketing Strategy

To effectively manage customer/user relationships throughout the project lifecycle, it's essential to establish diverse communication channels such as email, phone, and chat, ensuring regular updates on project progress. Implementing feedback mechanisms like surveys and user interviews helps gather insights, prioritizing user feedback for enhancements and fixes. Comprehensive support through dedicated teams and knowledge bases ensures prompt resolutions and high user satisfaction. Proactive engagement strategies, including personalized communication and targeted campaigns, foster user engagement and loyalty. Continuous monitoring of user interactions and feedback facilitates iterative improvements, fostering a culture of learning and innovation. Offering training resources and educational materials empowers users, promoting adoption and proficiency. Lastly, cultivating strong relationships through transparency, empathy, and appreciation fosters a sense of community and recognition among users, contributing to long-term success (Wagh, et al., 2024).

5. Conclusion

This report critically evaluated the development of a data science product designed for customer churn in the telecommunications industry. Learning from churn losses, it's evident that proactive customer interaction is essential for retention. The project showed that solving real-world business challenges using open-source data unlocks valuable insights through impactful visual data tools. The product, an interactive "Telecom Churn Dashboard", leverages Python for data processing, modeling, and analysis, ultimately presented through Power BI visualizations. The development process encompassed project management, software development, and user interface design.

Key findings include:

- Customer churn is a significant challenge in the telecom sector, necessitating proactive customer retention strategies.
- Data science techniques, coupled with open-source data, can provide valuable insights for mitigating churn.
- A well-designed dashboard with interactive visualizations empowers stakeholders to make informed decisions.

Future considerations for the project involve:

- Expanding the data size to potentially improve model performance.
- Exploring additional churn factors and incorporating them into the analysis.
- Integrating the dashboard with existing CRM systems for seamless data flow.

By effectively addressing customer churn, telecom companies can enhance customer satisfaction, minimize revenue losses, and gain a competitive edge in the marketplace. This project demonstrates the power of data science in tackling real-world business challenges and paves the way for further exploration and innovation in customer retention strategies.

6. References

- Anon., n.d. *Data Protection: The Data Protection Act*. [Online]
Available at: <https://www.gov.uk/data-protection>
[Accessed 2024].
- Cybiant, 2020. *Predicting Customer Churn in Power BI*. [Online]
Available at: <https://www.cybiant.com/knowledge/predicting-customer-churn-in-power-bi/>
[Accessed 2024].
- Kolomiiets, A., Mezentseva, O. & Kolesnikova, K., 2021. Customer churn prediction in the software by subscription models it business using machine learning methods. *In CEUR Workshop Proc*, Volume 3039, pp. 119-128. Available at: <http://star.informatik.rwth-aachen.de/Publications/CEUR-WS/Vol-3039/paper49.pdf>
- Ledro, C., Nosella, A. & Pozza, I. D., 2023. Integration of AI in CRM: Challenges and guidelines. *Journal of Open Innovation: Technology, Market, and Complexity*, 4(2199-8531), p. 100151. Available at: <https://doi.org/10.1016/j.joitmc.2023.100151>
- Pejić Bach, M., Pivar, J. & Jaković, B., 2021. Churn Management in Telecommunications: Hybrid Approach Using Cluster Analysis and Decision Trees. *Journal of Risk and Financial Management*, 14(11). Available at: <https://doi.org/10.3390/jrfm14110544>
- Saha, S. et al., 2024. ChurnNet: Deep Learning Enhanced Customer Churn Prediction in Telecommunication Industry. *IEEE Access*, Volume 12, pp. 4471-4484. Available at: <https://doi.org/10.1109/ACCESS.2024.3349950>
- Wagh, Sharmila K., A. & Aishwarya A., W. K. S. P. J. R. S. P. S., 2024. Customer churn prediction in the telecom sector using machine learning techniques. *Results in Control and Optimization*, Volume 14. Available at: <https://doi.org/10.1016/j.rico.2023.100342>
- Yang, P., 2023. Data Visualization and Prediction for Telecom Customer Churn. *Highlights in Science, Engineering and Technology*, Volume 39, pp. 1080-1085. Available at: <https://doi.org/10.54097/hset.v39i.6711>
- Zhang, T., Moro, S. & Ramos, R. F., 2022. A Data-Driven Approach to Improve Customer Churn Prediction Based on Telecom Customer Segmentation. *Future Internet*, 14(3). Available at: <https://doi.org/10.3390/fi14030094>
- Zimal, et al., 2023. Customer Churn Prediction Using Machine Learning. *International Journal for Research in Applied Science & Engineering Technology (IJRASET)*, 11(2). Available at: https://www.academia.edu/download/100216368/Customer_Churn_Prediction_Using_Machine_Learning.pdf

7. Appendix

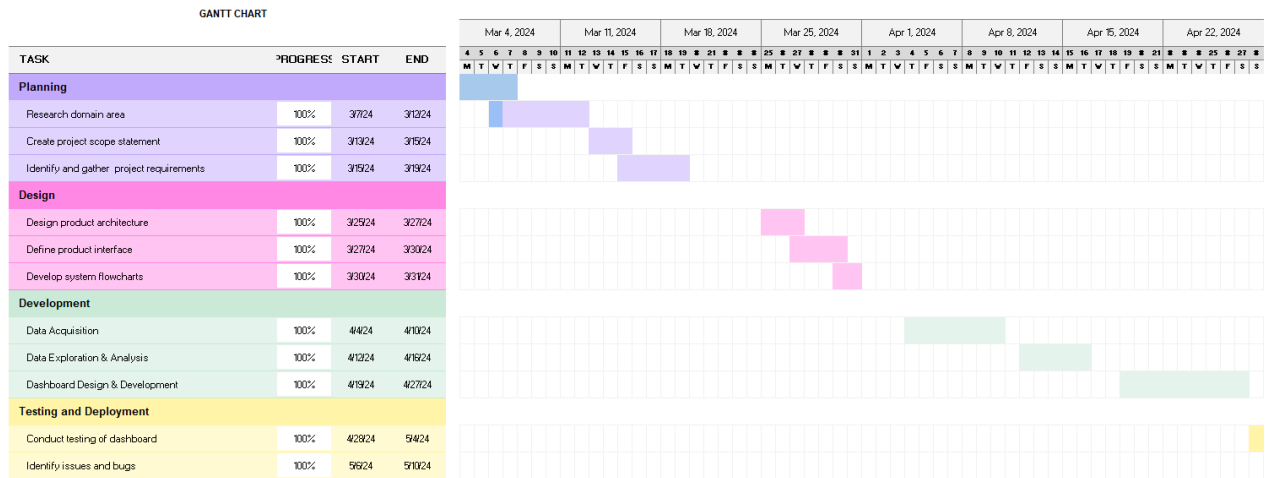


Figure 6: Project Gantt Chart



Figure 7: Customer Risk Analysis Dashboard before parameter selection



Figure 8: Customer Risk Analysis after parameters selection

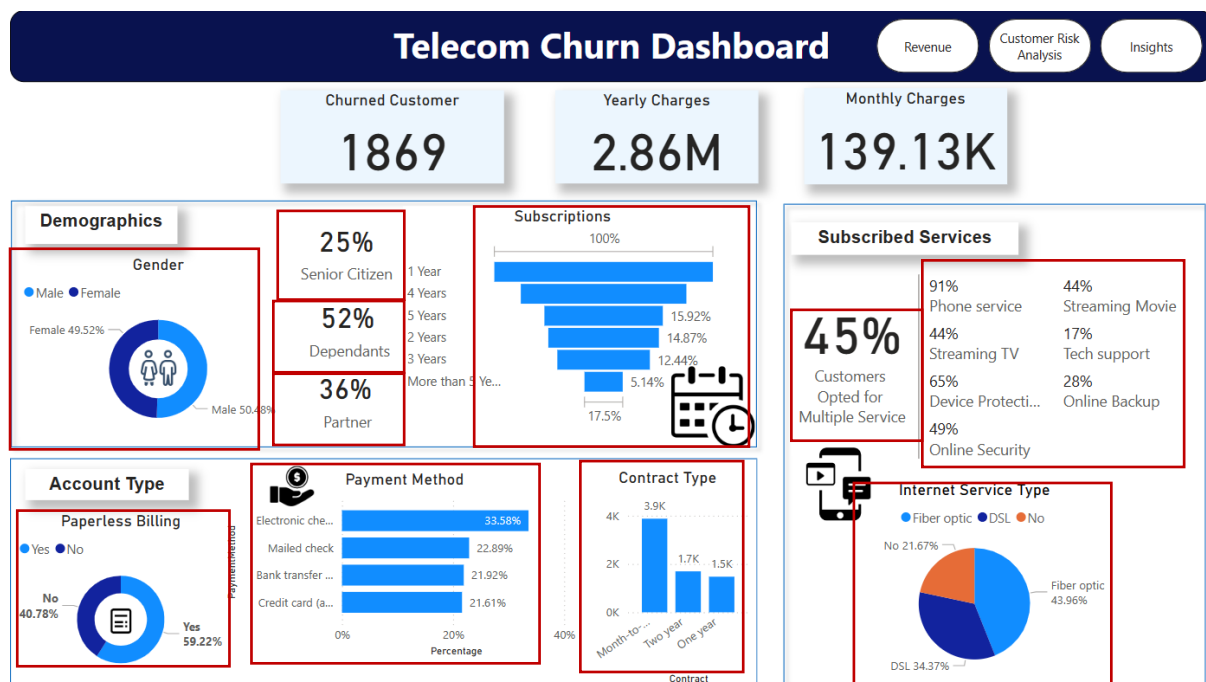


Figure 9: Telecom Churn Dashboard

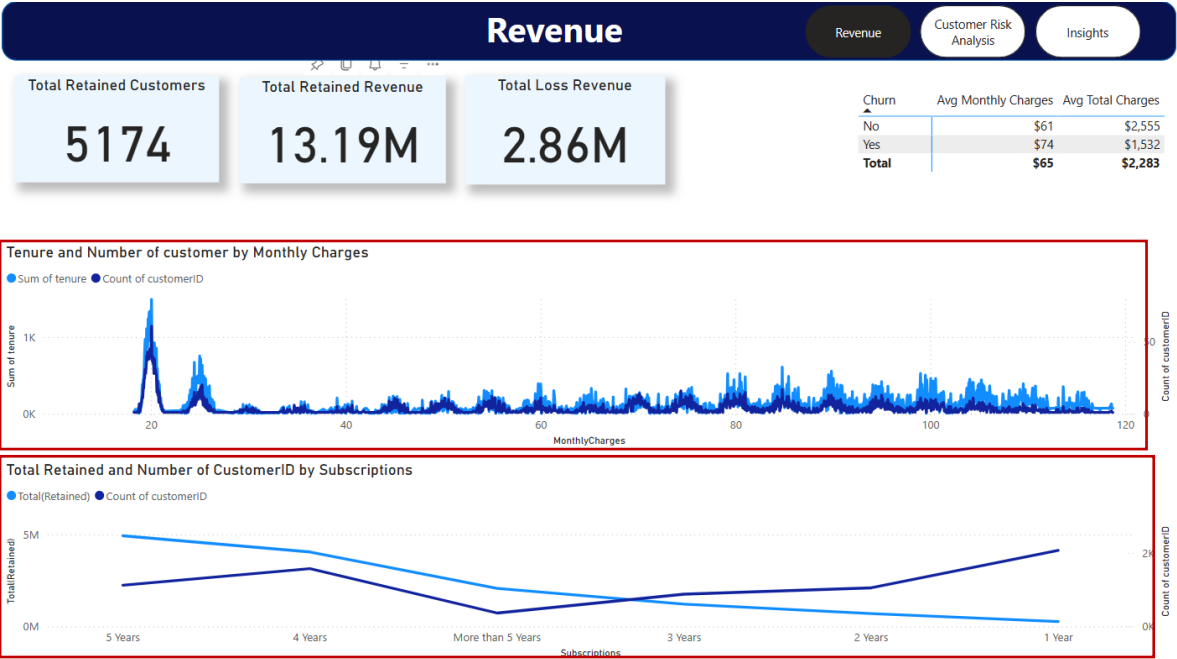


Figure 10: Revenue